EXPERIMENT 2

Write the Python program to solve the 8-Queens problem

Register Number: **192511013**

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Aim

To write a Python program that places 8 queens on an 8x8 chessboard so that no two queens attack each other (no two share the same row, column or diagonal), using the backtracking technique.

Algorithm

Step 1: Represent the board as a list where board[row] stores the column of the queen placed in that row. One queen per row is guaranteed by this representation.

Step 2: Start with row = 0.

Step 3: If row = 8, all queens are placed safely. Store the solution and backtrack to find more.

Step 4: For each column from 0 to 7 in the current row, check whether placing a queen there is safe: no earlier queen is in the same column or on the same diagonal (|col difference| = |row difference|).

Step 5: If the position is safe, place the queen and recursively solve for row + 1.

Step 6: If the recursive call returns (no further placement possible or a solution recorded), remove the queen and try the next column. This is the backtracking step.

Step 7: After all columns are tried, return to the previous row. Finally, print the total number of solutions and display the solutions as boards.

192511013

Program

```
N = 8

def is_safe(board, row, col):
    for r in range(row):
        c = board[r]
        if c == col or abs(c - col) == abs(r - row):
            return False
    return True

def solve(board, row, solutions):
    if row == N:
        solutions.append(board[:])
        return
    for col in range(N):
        if is_safe(board, row, col):
            board[row] = col
            solve(board, row + 1, solutions)
            board[row] = -1

def print_board(board):
    for r in range(N):
        print(' '.join('Q' if board[r] == c else '.' for c in range(N)))
    print()

if __name__ == "__main__":
    solutions = []
    solve([-1] * N, 0, solutions)
    print("Total number of solutions:", len(solutions))
    print("\nFirst solution (column of queen in each row):", solutions[0], "\n")
    print_board(solutions[0])
    print("Second solution:", solutions[1], "\n")
    print_board(solutions[1])
```

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Output

```
Total number of solutions: 92

First solution (column of queen in each row): [0, 4, 7, 5, 2, 6, 1, 3]

Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .

Second solution: [0, 5, 7, 2, 6, 3, 1, 4]

Q . . . . . . .
. . . . . Q . .
. . . . . . . Q
. . Q . . . . .
. . . . . . Q .
. . . Q . . . .
. Q . . . . . .
. . . . Q . . .
```

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